PHIL 133 · MAKING MINDS · FALL 2026

Midterm Prep

Thursday, November 5 · 12:30–1:45pm · in class

To prepare for the exam, make sure that you can solve the following problems with just pencil and paper.

Any circuit you solve can be boxed and used later.

There will be questions on the exam that are not on this prep sheet.

Combinatorial Circuits CCs

BASIC CIRCUITS

- ☐ XOR
- ☐ Always 1
- ☐ Half adder

TALLY

- ☐ +1 [0–4] T
- ☐ +2 [0–4] T
- ☐ +3 [0–4] T

BINARY

- ☐ +1 [0–15] B
- ☐ +2 [0–15] B
- ☐ +3 [0–15] B
- ☐ $2x$ [0–15] B
- ☐ $4x$ [0–15] B
- ☐ $x + y$ [0–15] B

SIMPLE COMBINATIONS

- ☐ $2x + 1$ [0–15] B
- ☐ $2(x + 1)$ [0–15] B

Sequential Circuits SCs

BASIC CIRCUITS

- ☐ Trip-wire
- ☐ ON/OFF switch

TALLY

- ☐ +1 T
- ☐ +2 T
- ☐ +3 T

BINARY

- ☐ +1 B
- ☐ +2 B
- ☐ +3 B
- ☐ $2x$ B
- ☐ $4x$ B
- ☐ $x + y$ B

COMBINATIONS: ONE VARIABLE

- ☐ $2x + 1$ B
- ☐ $2(x + 1)$ B
- ☐ $3x$ B
- ☐ $3x + 2$ B
- ☐ $4(x + 2)$ B
- ☐ $3(2x + 2)$ B

COMBINATIONS: TWO VARIABLES

- ☐ $x + y + 1$ B
- ☐ $2x + y$ B
- ☐ $2(x + y)$ B
- ☐ $3x + 2y$ B